3. Gibberellins promote the **production of male flowers** in monoecious plants (Cucurbits).
4. Gibberellins **break dormancy** of potato tubers.
5. Gibberellins are efficient than auxins in inducing the formation of seedless fruit - **Parthenocarpic fruits** (Development of fruits without fertilization) e.g. Tomato.

16.1.4 Abscisic Acid

Abscisic acid (ABA) is a **growth inhibitor** which **regulates abscission** and **dormancy**. It increases tolerance of plants to various kinds of stress. So, it is also called as **stress hormone**. It is found in the chloroplast of plants.

Physiological effects of abscisic acid

1. ABA promotes the process of **abscission** (separation of leaves, flowers and fruits from the branch).
2. During water stress and drought conditions ABA **causes stomatal closure**.

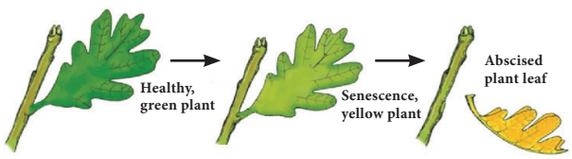


Figure 16.5 Senescence and abscission

Activity

Place two or three unripe tomatoes in a brown paper bag with a ripe bananas and roll the top closed. In another bag place two or three unripe tomatoes only and roll the top closed, Observe what happens to the tomatoes? Why?

As the banana continues to ripen in the first bag, it produces ethylene gas. The gas trapped in the bag will cause tomatoes to ripen.The tomatoes remain unripe in the second bag.

3. ABA **promotes senescence** in leaves by causing loss of chlorophyll.
4. ABA **induces bud dormancy** towards the approach of winter in trees like birch.
5. ABA is a powerful **inhibitor of lateral bud growth** in tomato.

16.1.5 Ethylene

Ethylene is a **gaseous plant hormone**. It is a **growth inhibitor**. It is mainly concerned with maturation and ripening of fruits. Maximum synthesis of ethylene occurs during ripening of fruits like apples, bananas and melons

Physiological effects of ethylene

1. Ethylene promotes the **ripening of fruits**. e.g. Tomato, Apple, Mango, Banana, etc.



Figure 16.6 Ethylene

2. Ethylene **inhibits** the **elongation** of stem and root in dicots.
3. Ethylene hastens the **senescence** of leaves and flowers.
4. Ethylene stimulates **formation of abscission zone** in leaves, flowers and fruits. This leads to premature shedding.
5. Ethylene **breaks the dormancy** of buds, seeds and storage organs.

16.2 Human Endocrine Glands

Endocrine glands in animals possess a versatile communication system to coordinate biological functions. Exocrine glands and endocrine glands are two kinds of glands found in animals. Endocrine glands are found in different regions of the body of animals as well as human beings. These glands are called **ductless**

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